Exact Fault-Robust Stretch of Period-Three Temporal Cycles

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Abstract

We study an undirected cycle whose edges are available at one assigned phase in each period of length three. After a labeling is chosen, at most one edge phase may be replaced arbitrarily, with the replacement recurring in every period. We minimize the worst resulting ratio of fastest strict temporal-path duration to static distance, over all ordered vertex pairs. The exact value is 1 for $n = 3$, $3/2$ for $n = 4$, 2 for $n = 5, 7$ and for multiples of three at least six, and $7/3$ for all remaining $n \geq 8$. The proof uses a forward–backward path identity, a three-edge obstruction, and explicit constructions, including a seven-cycle exception. Exact enumeration and a separate clock-time implementation support the proof. An external AI review reports proof and computation PASS. Historical novelty remains inconclusive; this is an unrefereed public preprint without a priority claim.

1 Model and result

Let C_n have vertices $0, \ldots, n-1$ and edges $e_i = \{i, i+1\}$, with indices modulo n and $n \geq 3$. A labeling $\lambda \in \mathbb{Z}_3^n$ makes e_i available at all nonnegative integer times congruent to λ_i modulo 3. Traversal times along a temporal path must be strictly increasing. Following the fastest-path convention of stretched temporal graph realization (STGR) [1], define

$$T(a_1, \ldots, a_m) = 1 + \sum_{j=1}^{m-1} w(a_j, a_{j+1}), \qquad w(a, b) = 1 + ((b - a - 1) \bmod 3). \tag{1}$$

Here a path's duration is its last traversal time minus its first traversal time plus one. The departure time can be chosen freely; waiting before the first traversal is excluded. For an ordered pair (u, v), $T_\lambda(u, v)$ is the smaller duration of the two simple directed arcs from u to v. This suffices because these are the only simple paths in a cycle. Allowing walks cannot improve the minimum: delete a closed subwalk and wait at its initial vertex until the original onward traversal. If deletion removes an initial or final segment, duration can only decrease.

Write $d(u, v)$ for static cycle distance and set

$$S(\lambda) = \max_{u \neq v} \frac{T_\lambda(u, v)}{d(u, v)}, \quad R(\lambda) = \max_{d_{\mathrm{H}}(\lambda, \mu) \leq 1} S(\mu), \quad \beta_3(C_n) = \min_{\lambda \in \mathbb{Z}_3^n} R(\lambda). \tag{2}$$

The Hamming distance counts edges whose phases differ. The fault is fixed before routes and departure times are chosen, is persistent across periods, and includes the option of no change. The static graph and its distance denominator remain fixed.

Theorem 1 (Exact one-phase-fault classification)**.** *For every integer $n \geq 3$,*

$$\beta_3(C_n) = \begin{cases} 1, & n = 3, \\ 3/2, & n = 4, \\ 2, & n = 5, 7, \text{ or } n \geq 6 \text{ with } 3 \mid n, \\ 7/3, & n \geq 8 \text{ with } 3 \nmid n. \end{cases}$$

These cases are disjoint and cover all $n \geq 3$. The numerical statement is the one carried forward from the research handoff. We provide a self-contained proof; no result from the author's fault-free cycle note [3] is needed to establish it.

2 Path identities and lower bounds

Lemma 1 (Two directions)**.** *For a phase word $a_1, \ldots, a_m$, let F and B denote its forward and reverse durations, and let q count indices j with $a_j = a_{j+1}$. Then*

$$F + B = 3m - 1 + 3q \geq 3m - 1. \tag{3}$$

Proof. If $a \neq b$, then $w(a, b) + w(b, a) = 3$; if $a = b$, the sum is 6. Add over the $m - 1$ transitions and include the initial 1 in each duration. □

Lemma 2 (Effect of one replacement)**.** *Replacing an endpoint phase of a path with at least two edges can increase its duration by at most 2. Replacing an internal phase can increase duration by at most 3.*

Proof. An endpoint changes one wait, which lies in $\{1, 2, 3\}$. An internal replacement changes a sum $w(a, b) + w(b, c)$. Both its old and new values lie in $[2, 6]$ and are congruent to $c - a$ modulo 3, so their difference is at most 3. A one-edge path always has duration 1. □

Lemma 3 (Universal small-distance lower bounds)**.** *Every labeling of C_4 has robust stretch at least $3/2$. Every labeling of C_n with $n \geq 5$ has robust stretch at least 2.*

Proof. Take any two consecutive edges and, if necessary, replace one phase to make their phases equal. The resulting two-edge arc takes time 4 in both orientations. Its endpoints have static distance 2 for $n \geq 4$. The complementary arc has $m = n - 2$ edges. By Lemma 1, at least one of its two orientations takes time at least $\lceil (3m - 1)/2 \rceil$. Choose the ordered endpoints so this slower orientation is used on the complement. For $n = 4$ this quantity is 3, so both available arcs take at least 3. For $n \geq 5$ it is at least 4, so both arcs take at least 4. Dividing by 2 gives the claims. If the chosen phases were already equal, the no-fault option is sufficient. □

Lemma 4 (Three-edge obstruction)**.** *If $n \geq 8$ and $3 \nmid n$, then every labeling of C_n has robust stretch at least $7/3$.*

Proof. If every cyclic triple of consecutive edge phases is pairwise distinct, then $\lambda_{i+3} = \lambda_i$ for every i: both λ_i and λ_{i+3} must be the third element of $\mathbb{Z}_3 \setminus \{\lambda_{i+1}, \lambda_{i+2}\}$. Such a cyclic word is a repetition of a permutation of $0, 1, 2$ and has length divisible by 3. Thus some triple has a repeated phase. At most one replacement makes all three phases equal (possibly no replacement if already uniform).

The resulting three-edge arc takes duration 7 in either orientation. Its complementary arc has $m = n - 3 \geq 5$ edges and, by Lemma 1, has an orientation with duration at least $\lceil (3m - 1)/2 \rceil \geq 7$. Choose the ordered endpoints for that orientation of the complement. Both routes then take at least 7, while their static distance is 3 because $n \geq 8$. □

The restriction $n \geq 8$ matters: the complementary arc on C_7 has only four edges, for which Lemma 1 guarantees only a duration of 6 in one orientation.

3 Upper bounds for the general constructions

Use the phase word

$$\lambda_i = i \bmod 3, \qquad 0 \leq i < n. \tag{4}$$

In clockwise order, every transition has wait 1 except possibly the wrap from e_{n-1} to e_0. In reverse order regular waits are 2. Every simple arc contains the wrap transition at most once. Its two directional waits are

$n \bmod 3$	Clockwise wrap wait	Reverse wrap wait
0	1	2
1	3	3
2	2	1

Lemma 5 (Multiples of three)**.** *If $3 \mid n$, the labeling (4) has robust stretch at most 2.*

Proof. Fix a fault and an ordered vertex pair, and choose a shortest static arc of length $m = d(u, v)$. For $m = 1$, duration is 1. For $m = 2$, any phase word has duration at most 4.

Suppose $m \geq 3$. A clockwise arc originally has duration m. Lemma 2 bounds its post-fault duration by $m + 3 \leq 2m$. A reverse arc originally has duration $2m - 1$. If the modified edge is an endpoint of this arc, its single affected wait originally equals 2 and can increase by at most 1, giving duration at most $2m$. If it is internal, the neighboring phases form a descending permutation of the three phases. The two affected waits originally sum to 4 and, after either nontrivial replacement of the middle phase, become $(3, 1)$ or $(1, 3)$, still summing to 4. If the fault is outside the arc, its duration is unchanged. Thus this arc always has duration at most $2m$. □

Lemma 6 (Other cycle lengths)**.** *For every $n \geq 3$, the labeling (4) has robust stretch at most $7/3$.*

Proof. Again take a shortest static arc of length m. For $m = 1$ its duration is 1; for $m = 2$ it is at most 4; for $m = 3$ it is at most 7, regardless of the labeling.

For a clockwise arc with $m \geq 4$, the original duration is at most $m + 2$, because its only exceptional transition exceeds the regular wait by at most 2. By Lemma 2 its post-fault duration is at most

$$m + 5 \leq 2m + 1 \leq \tfrac{7}{3}m.$$

For a reverse arc, we show that its post-fault duration is at most $2m + 1$. If the wrap transition is absent, all waits are originally 2; an internal replacement preserves the sum 4 of its two affected waits, while an endpoint replacement increases a wait by at most 1. The bound is then $2m$.

If the wrap is present and $n \equiv 1 \pmod 3$, its wait is 3 and the original duration is $2m$. An internal replacement away from it again preserves a sum 4. One touching it changes a pair $(3, 2)$ or $(2, 3)$ with sum 5 and distinct outer phases. Every possible new sum has the same residue modulo 3 and lies in $[2, 6]$, so it is at most 5. An endpoint replacement increases an ordinary wait by at most 1 and cannot increase the exceptional wait 3. Hence the new duration is at most $2m + 1$.

If the wrap is present and $n \equiv 2 \pmod 3$, its wait is 1 and the original duration is $2m - 2$. An internal replacement away from it preserves sum 4. One touching it changes a pair $(1, 2)$ or $(2, 1)$ of sum 3, which can increase to at most 6. An endpoint replacement increases a wait by at most 2. Thus the new duration is again at most $2m + 1$. When $3 \mid n$ there is no exceptional wait, so the first case applies.

Since $2m + 1 \leq 7m/3$ for $m \geq 3$, all the required bounds hold. The fastest route is no slower than the particular shortest static arc just bounded. □

4 Small cycles and the seven-cycle exception

For $n = 3$, every pair is adjacent and always has duration 1. For $n = 4$, use 0101. For any opposite pair, the two two-edge arcs are edge-disjoint, so at least one is unchanged by a single fault. Its original consecutive phases differ, giving duration at most 3 in either required orientation. Adjacent pairs still have duration 1. Hence robust stretch is at most 3/2. For $n = 5$, every pair has distance at most 2, and the universal two-edge bound $T \leq 4$ gives robust stretch at most 2 for every labeling.

For $n = 7$, use the specified witness

$$\lambda = (0, 1, 0, 1, 2, 0, 2). \tag{5}$$

Distances 1 and 2 satisfy the target bound 2 automatically. Consider an ordered pair at distance 3, and its unique three-edge shortest static arc. A three-phase word that is not uniform has duration at most 6: attaining 7 requires both waits to be 3, hence all phases equal. If its original phases are all distinct, one replacement cannot make them uniform. Otherwise its starting edge is one of $0, 1, 4, 5$, as shown in Table 1. If a fault makes this triple uniform, it must lie on the triple, and the complementary four-edge arc is unchanged. The table gives its duration in both orientations, each at most 6. Thus every ordered pair at distance 3 still has a route of duration at most 6, proving robust stretch at most 2.

Starting edge i	Triple	Complement in clockwise order	F	B
0	010	1202	5	6
1	101	2020	5	6
4	202	0101	5	6
5	020	1012	5	6

Table 1: The four non-rainbow cyclic triples of (5). The complement uses edges $e_{i+3}, \ldots, e_{i+6}$. Its forward and backward durations are computed from (1).

Proof of Theorem 1. Every temporal path has at least the static distance in edge traversals, so stretch is always at least 1. The small-cycle constructions above settle $n = 3, 4, 5, 7$ together with Lemma 3. For multiples of three at least six, combine Lemmas 3 and 5. For all remaining $n \geq 8$, combine Lemmas 4 and 6. These exhaust all cases. □

5 Reproducibility and computational scope

The bundle contains two implementations using exact integers and rational comparisons. The arc evaluator computes durations from prefix sums of (1). The clock-time oracle builds the undirected graph, advances an integer clock until the next edge availability, and runs a time-dependent Dijkstra algorithm for each source and each starting threshold $0, 1, 2$. Taking the smallest duration across these thresholds realizes free departure. Initial waiting is charged in each threshold run, but a threshold equal to the first edge phase charges none; thus the minimum has precisely the intended duration. The clock module imports neither the arc evaluator nor the proposed theorem values.

Adding the same element of $\mathbb{Z}_3$ to all phases preserves every duration and maps each radius-one Hamming ball bijectively onto the shifted ball. Therefore $\lambda_0 = 0$ is an exact normalization, leaving 3^{n-1} labelings. For each labeling, robust evaluation includes its original score and all $2n$ nontrivial one-edge changes. The optimization enumerations do not use the theorem to discard candidates; the theorem is compared only after the minimum is computed.

n	Normalized labelings	Optimal labelings	Computed minimum	Clock recheck
3	9	9	1	all
4	27	6	3/2	all
5	81	81	2	all
6	243	164	2	all
7	729	14	2	all
8	2,187	2,050	7/3	all
9	6,561	2	2	all
10	19,683	2,702	7/3	selected witnesses
11	59,049	2,706	7/3	selected witnesses

Table 2: Fresh enumeration in Python 3.12.14 on 24 October 2026. Counts distinguish labeled edge positions and normalized phases; rotations, reflections, and phase permutations are not quotiented further.

The full run checked 88,569 normalized assignments for $3 \le n \le 11$. The two implementations agreed on both ordinary and robust scores for every one of the 9,837 normalized assignments with $3 \le n \le 9$. The explicit construction, including its no-fault case and all $2n$ faults, was checked for every $3 \le n \le 36$. For $n = 3, \ldots, 12, 17, 24, 31, 36$, all these faulted construction distance matrices were compared against the clock oracle, totaling 198,568 ordered-pair entries. These selected-witness comparisons include $n = 10, 11$; there is no full clock enumeration for those two sizes.

In addition, all 9,840 phase words of lengths 1 through 8 satisfied the exact identity (3); 147,624 nontrivial single-phase replacements satisfied Lemma 2. The special C_7 file lists the no-fault case and all 14 nontrivial faults, and its complement table is separately reproduced. No counterexample or implementation disagreement was found in these stated finite checks. These checks support the analytic proof, but do not prove its infinite quantification.

Run `python check_hashes.py` and `python verify.py` from the extracted directory. No third-party Python package, external API, paid service, or network access is needed. The submitted run used Linux with Python 3.12.14; Windows 10 execution has not been performed here. Windows-compatible commands and an explicit output directory are provided in the README. Re-execution writes fresh evidence separately and does not overwrite the shipped evidence.

6 Related work and claim limits

Mertzios et al. [1] introduced STGR and its local-search relabeling neighborhood. The JCSS author manuscript uses the acronym SPTGR. Its Section 2 has the same strict fastest-duration/static-distance objective; Section 7 asks whether some nearby labeling improves stretch. Our objective instead minimizes over centers the maximum inside a radius-one Hamming ball. The temporal model, modification operation and general robustness concept are prior art. The familiar period-three bound $3 - 2/d$ already gives 7/3 at static distance three; this fraction alone is not a novelty claim.

Meusel et al. [2] study ordinary STGR realizability, hardness and possible stretch values. Their Theorem 3.3 also yields a 7/3 existence lower bound for an ordinary general-graph family after parameter substitution. That statement does not itself identify the robust optimum on each cycle. The author's prior fault-free cycle note [3] uses the same duration convention but minimizes ordinary stretch. The present proof is self-contained.

Bilò et al. [4] study blackout-tolerant and edge-fault-tolerant temporal spanners. Their indexed

Section 2 uses nondecreasing labels and minimum hop distance, comparing a subgraph against the original temporal graph after a deletion fault. No.37 uses recurring phase replacement and elapsed duration against fixed static distance. Enright et al. [5] study finite appearance-wise time perturbations, including fastest eccentricity. Their allowed modifications and existential improvement objective differ from a persistent periodic edge error and worst-case all-pairs design. Füchsle et al. [6, 7] study delay-robust routes and connections, including paths that adapt after delays are announced. Universal disruption followed by an adapting path is consequently not a new general quantifier pattern.

The residual comparison obtained the complete 58-page author-hosted JCSS manuscript and examined the relevant model and local-search sections; the MFCS and SAND 2026 primary texts, the author's No.27 note, and the perturbation and delay-model definitions were also checked. Bilò's full journal PDF remained inaccessible; its definitions were examined through indexed primary text, with its broader fault coverage checked against the publisher abstract. No direct specialization yielding Theorem 1 was identified in the statements examined. This is not an exhaustive historical or reduction-theoretic non-overlap result. **REFERENCES remain PARTIAL and NOVELTY remains INCONCLUSIVE. No priority claim is made.**

Received external AI review. The author supplied a review reporting PROOF and COMPUTATION PASS on Linux/Python 3.13.17, with manual proof tracing, the 23/23 candidate hashes, independent C++ enumeration through $n = 16$, and construction checks through $n = 39$. Those additional C++ sources and raw logs were not supplied in this bundle and have not been rerun here. They remain reviewer-reported evidence. The archived author-side exhaustive range is $n = 3, \ldots, 11$. The reported 162,378 model-comparison count equals the phase-normalized count for $n = 3, \ldots, 8$, not the literal unnormalized count; the phase-shift symmetry accounts for the factor three. Formal proof-assistant verification and journal/conference peer review are not claimed.

Author release decision. The independent AI reviewer recommended PUBLICATION HOLD with novelty inconclusive. After receiving that report and the scoped literature comparison, the author explicitly instructed public release on 24 October 2026. This is a qualified, unrefereed preprint. The historical audit recommendation is retained in the evidence archive and is not represented as reviewer release approval. Research inquiries and earlier overlapping results are welcome at the contact on page 1.

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